

Customer, Risk and Growth Detailed Analysis

1. Introduction and scope

The brief asked for any three of four tasks. I chose to go deep on exploratory analysis, customer segmentation and the management dashboard, and to treat forecasting as a short outlook section rather than a full modelling exercise. The reasoning is practical: five of the six concerns raised by leadership (suspicious activity, onboarding costs, new-user retention, KYC drop-off, failed transactions, uneven segment value) are diagnostic questions, and only one is forward-looking.

No dataset was provided, so I generated one. Section 2 explains how it was built and, more importantly, how I kept the analysis honest despite having built the data myself. The rest of the report follows the order of the work: data quality first, then what the data looks like, then what it shows, then who our customers are, then what the next quarter may hold, and finally the dashboard through which management can explore all of it.

2. Data and method

The dataset simulates a Nepali digital wallet with six product lines: wallet top-up, peer-to-peer transfer, merchant QR payment, bill payment, bank transfer and loan repayment. It covers 24 months (August 2024 to August 2026), one million users and 25.3 million transactions, exceeding the minimum scale set in the brief. The generator encodes patterns known from real payment businesses rather than patterns convenient for my analysis: transaction amounts follow a different distribution per product (fixed top-up denominations, log-normal merchant and transfer amounts), activity clusters on weekdays and around payday, a KYC funnel loses users at each stage, an infrastructure incident degrades two payment rails for two weeks in March 2026, and a small group of accounts splits bank transfers to stay under the Rs 100,000 reporting threshold. I also planted realistic defects on purpose, including duplicate records, several distinct kinds of missing data, and per-user behavioural noise that blurs the boundaries between customer types.

Two safeguards keep the exercise honest. First, the true segment label of every user is written to a separate file that the analysis pipeline never reads; it is used exactly once, at the end, to grade the segmentation. Second, all results in this report were produced by code that would run unchanged on real transaction data of the same shape. Because the dataset is larger than the memory of the machine it was built on, both generation and analysis run in batches partitioned by the user, which is valid because every operation involved is local to a single user. Raw data is never overwritten; cleaning writes a separate layer.

One methodological point recurs throughout: the unit of analysis. Questions about events (failure rates, incident windows) are answered on the 25-million-row transaction table, while questions about people (segments, churn, funnel conversion) are answered on a one-million-row per-user feature table. Mixing the two grains silently weights every user-level statistic by transaction count, which is one of the commonest ways dashboards mislead.

3. Data quality

Before drawing conclusions I audited the data itself. The audit found 25,274 duplicate transaction records, exactly 0.100 percent of the total, consistent with occasional double capture at ingestion. Duplicates were removed into the clean layer used for everything that follows; left in place they would overstate volume and payment value. Every structural check passed: no duplicate users, no transactions attributed to unknown accounts, no transactions dated before the user signed up or after the snapshot date, no non-positive amounts, and no KYC verification recorded before its submission.

Check	Result
Duplicate transaction records	25,274 (0.100%), removed into clean layer
Duplicate users / orphan user references	0 / 0
Transactions before signup / after snapshot	0 / 0
Non-positive amounts	0
KYC completed before submitted	0

Table 1. Data quality checks on the raw extract.

Missing values needed classification before treatment, because most of them are not defects. The failure reason field is empty on 95.5 percent of rows for the simple reason that those transactions succeeded; among failed transactions it is always populated. The merchant identifier is absent on every peer-to-peer and bank transfer because those transactions have no merchant; its absence carries meaning and imputing it would fabricate data. Genuine gaps are small: the device field is missing at random on 3 percent of rows. Income bracket is unknown for 72 percent of users, which reflects how wallets actually onboard customers; I excluded it from modelling rather than guessing it.

4. Exploratory analysis

4.1 The shape of the data

Transaction amounts are not one distribution but a mixture. Top-ups sit on a handful of fixed denominations between Rs 50 and Rs 1,000, merchant and transfer amounts spread across a long right tail, and bank transfers reach two orders of magnitude higher. Averages are therefore misleading for this data, and the report relies on medians and percentiles throughout.

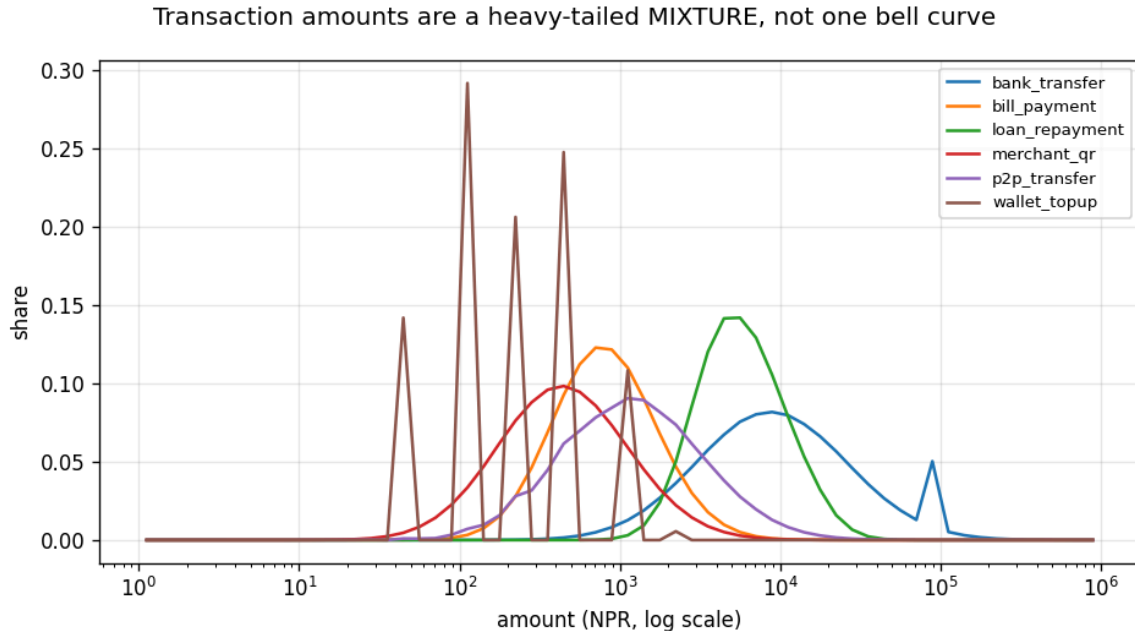


Figure 1. Amount distribution by product line, log scale. Each product has its own shape; no single curve describes them.

Activity per user is heavily skewed. A minority of users generates most of the volume, which is why a statistic computed per transaction and the same statistic computed per user can both be correct yet differ widely, and why this report always states which of the two it is reporting.

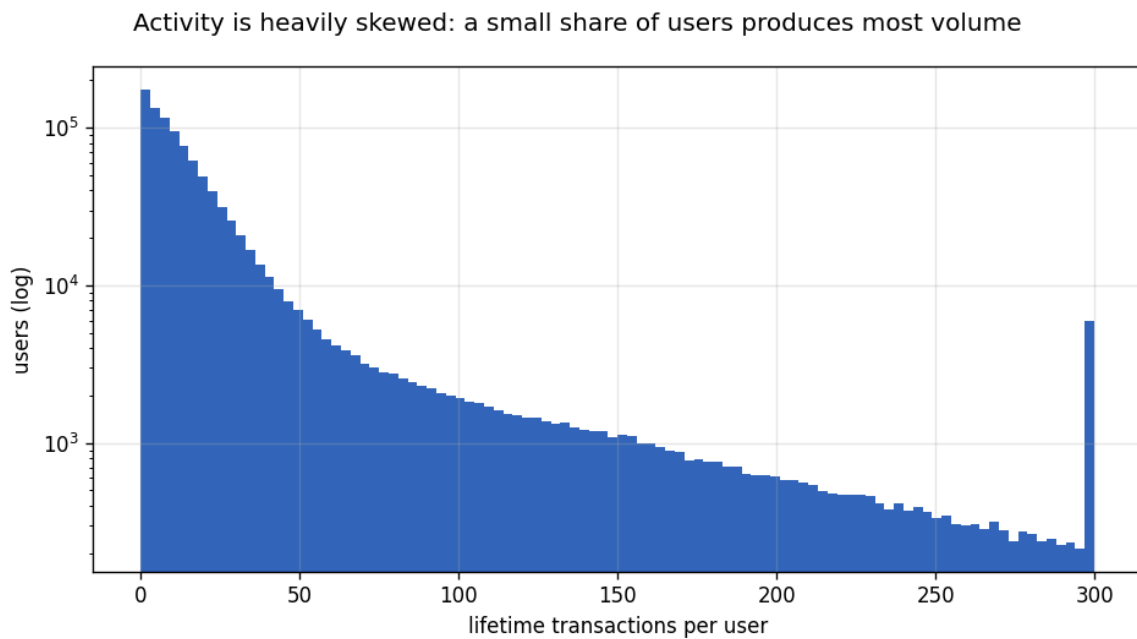


Figure 2. Lifetime transactions per user (log count axis). Most users transact little; a small group transacts constantly.

The days elapsed since each user's last transaction form two distinct groups: an engaged population seen within the last month or two, and a long-departed one. I place the churn threshold at 90 days, in the valley between the two humps, so that the definition of a churned user comes from the data rather than from convention. On that definition 19.5 percent of users have churned.

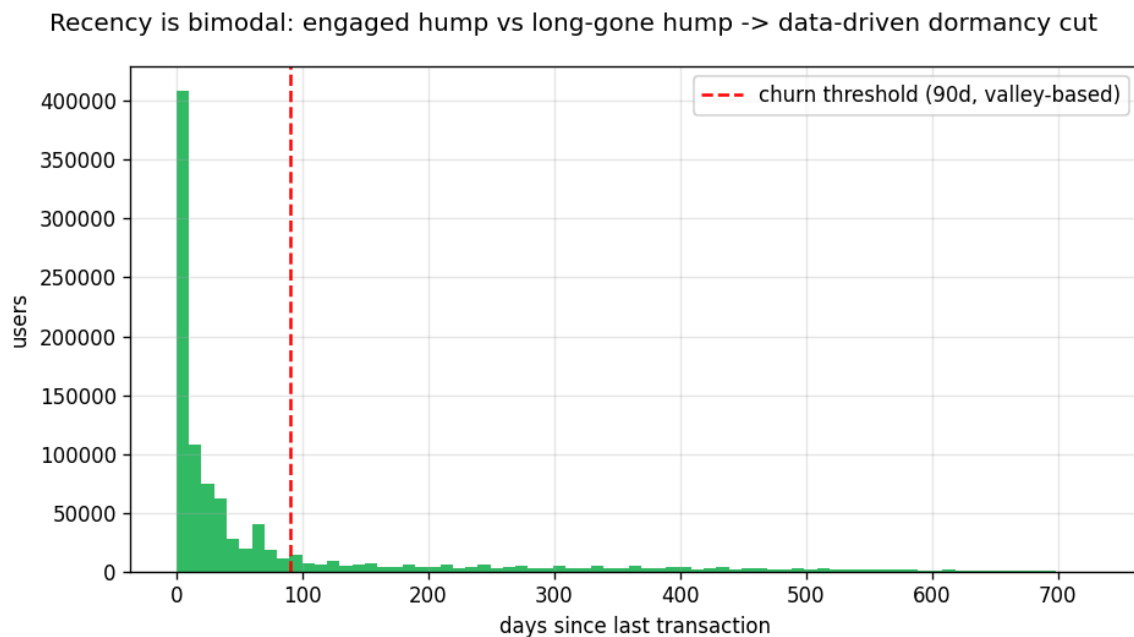


Figure 3. Days since last transaction. The 90-day churn threshold sits in the valley between the engaged and departed populations.

4.2 What the data shows

The most consequential result concerns the beginning of a customer's life. A user's very first transaction fails 13.8 percent of the time. The rate falls to 9.3 percent on the second transaction, 6.3 on the third, and settles near 3.9 percent from the fourth onward. If failures were spread evenly across a customer's lifetime this curve would be flat; instead the risk is concentrated precisely where the customer is least committed.

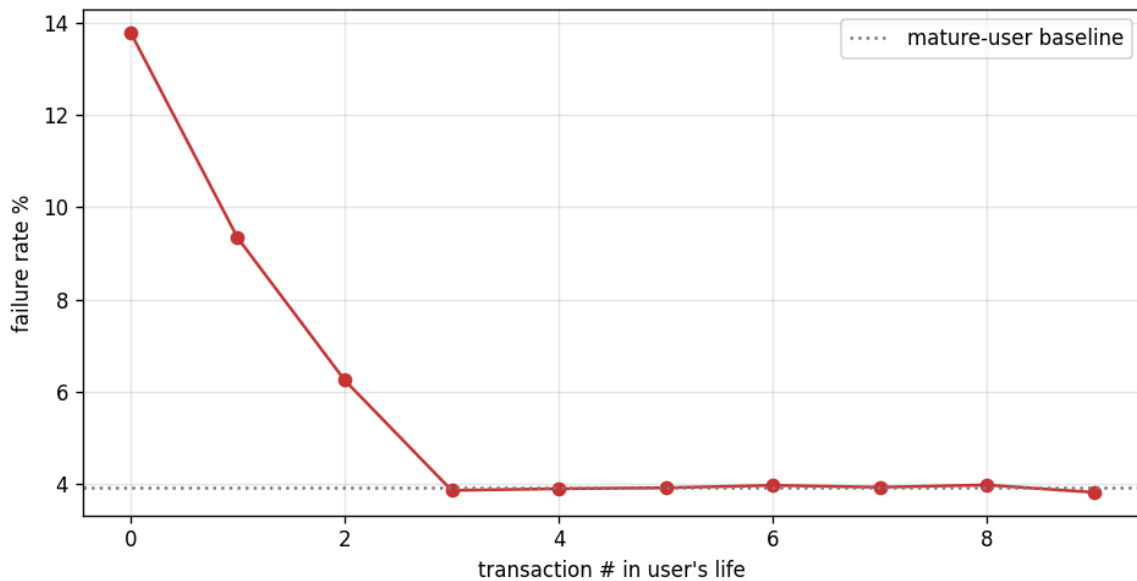


Figure 4. Failure rate by position of the transaction in the user's life. First transactions fail at roughly 3.5 times the mature baseline.

The commercial cost of that early unreliability is visible in retention. Among users whose first transaction succeeded, 95.8 percent went on to make a second one. Among users whose first transaction failed, only 74.5 percent did. That is a gap of 21 percentage points caused by a single bad first experience, and it links three of the concerns leadership raised (failed transactions, weak new-user retention, rising acquisition cost) into one engineering problem: money spent acquiring a user is written off at the moment their first payment fails. In this dataset the link is causal by construction; on production data I would confirm it with a cohort or survival analysis before committing engineering budget, but the pattern is exactly what that analysis would test.

A failed first transaction costs ~21 points of early retention

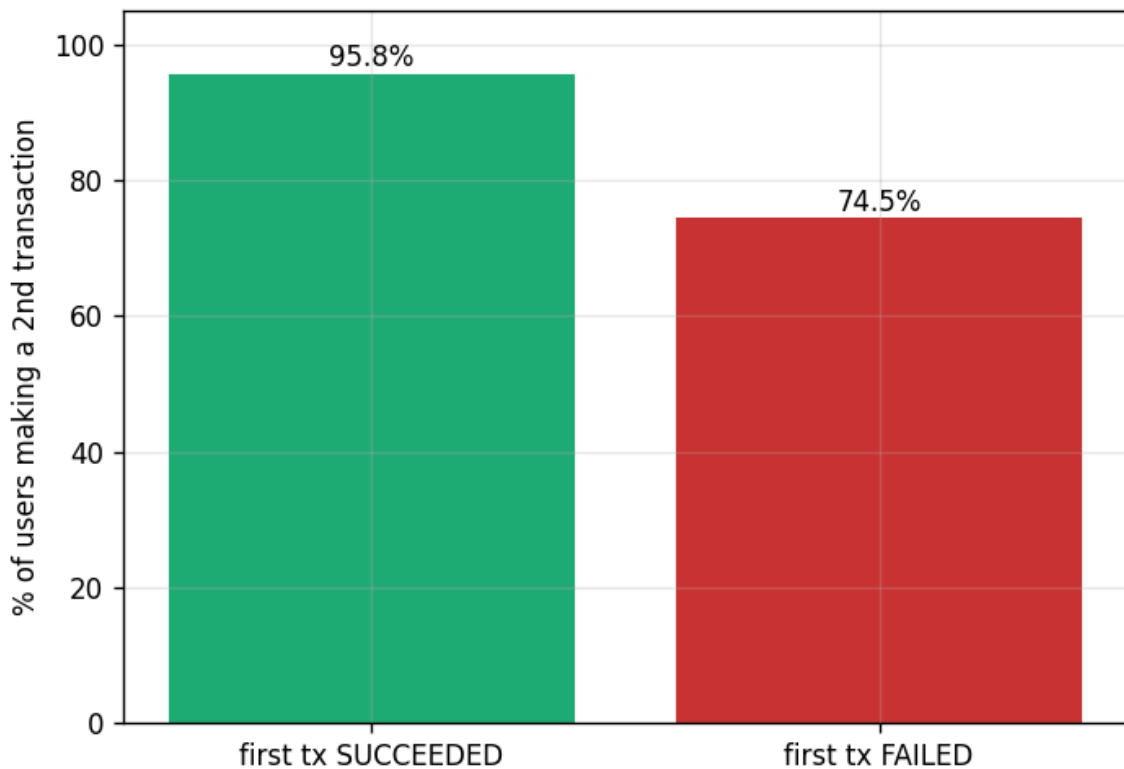


Figure 5. Share of users who make a second transaction, split by the outcome of their first. The 21-point gap is the price of a failed first payment.

Onboarding leaks earlier than that, at verification. Of one million signups, 786,286 started KYC and 713,093 completed it, so 21.4 percent never begin verification and a further 7.3 percent abandon or fail it. Notably, 96.7 percent of users transact at least once, more than are verified: transacting does not require completed KYC because unverified accounts operate under limits. That inversion is deliberate in the funnel below and is itself a compliance observation, since a share of platform activity necessarily flows through unverified accounts. Meanwhile 3.3 percent of signups never transact at all, an acquisition cost with zero return.

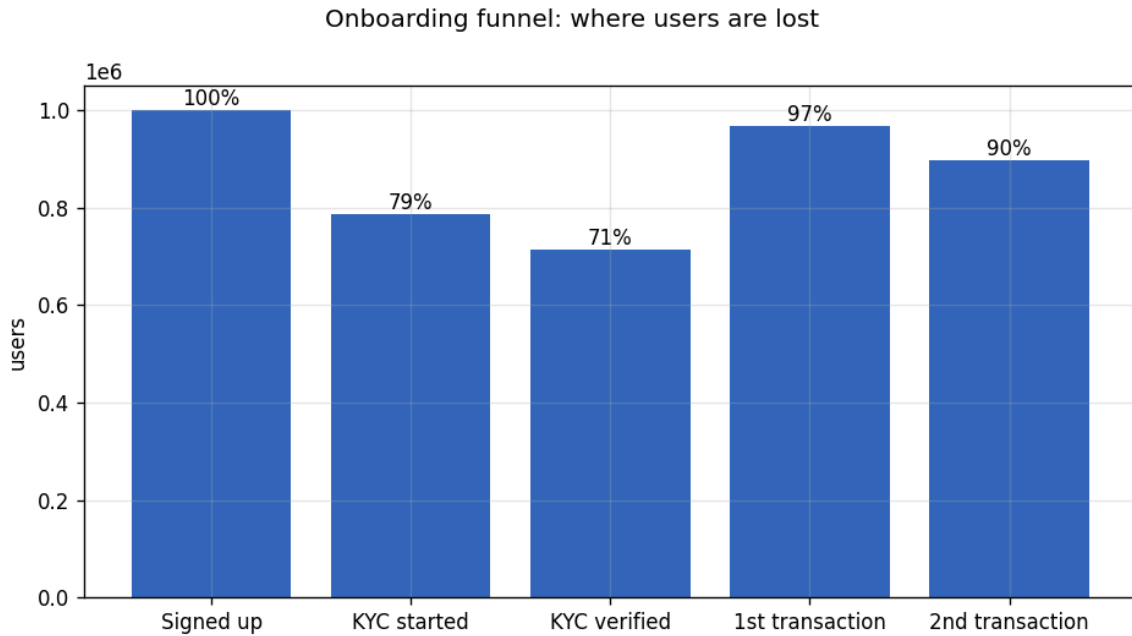


Figure 6. Onboarding funnel. Verification, not signup, is where acquisition spend leaks; transacting is possible before verification, under limits.

In March 2026 the failure rate spiked for two weeks. The spike is confined to the merchant QR and bank transfer rails; every other product line stays at its baseline, and amounts and volumes remain normal throughout. That pattern rules out both a fraud wave (which would not respect rail boundaries so cleanly) and a data collection outage (which would distort volumes as well), and points to a payment gateway incident. The affected rails and dates give engineering a precise scope for the incident review.

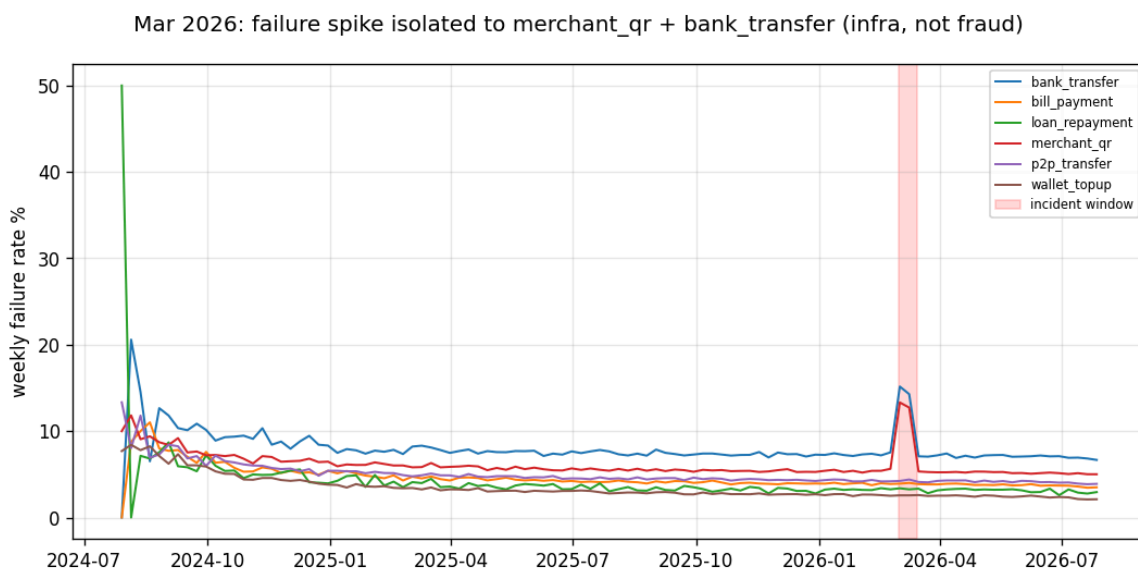


Figure 7. Weekly failure rate by product line. The March 2026 spike touches exactly two rails and nothing else.

Finally, bank transfer amounts show a distinct pile-up just below Rs 100,000. Splitting transfers to remain under a reporting threshold is a well-known money-laundering signature called structuring, and the accounts producing this pattern overlap heavily with the risk segment described in the next section. This is the kind of finding that would justify a suspicious-activity review in a real institution.

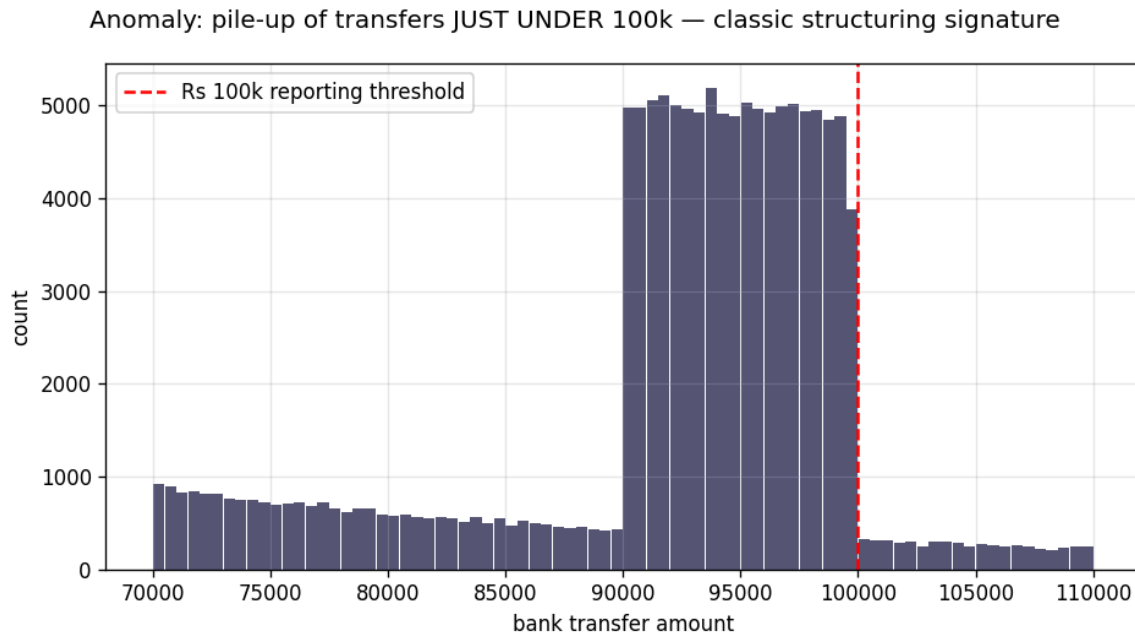


Figure 8. Bank transfer amounts between Rs 70,000 and 110,000. The concentration just under the Rs 100,000 threshold is a classic structuring signature.

4.3 Statistical checks

The headline retention result deserves a formal test rather than a visual comparison. Treating users whose first payment failed and users whose first payment succeeded as two independent samples, the difference in second-transaction rates is 21.3 percentage points (95.75 against 74.50 percent) with a 95 percent confidence interval of 21.0 to 21.5 points. A two-proportion z-test returns a statistic of 279; at a million users the sampling uncertainty is negligible and the p-value falls below any conventional threshold. The gap is a real, precisely measured difference, not an artefact of sample size.

Correlations among the customer features, measured with Spearman's rank coefficient because the variables are heavily skewed, tell a consistent story. The engagement variables move together: transaction frequency correlates with lifetime spend at 0.74, service diversity with spend at 0.84, and recency runs against frequency at minus 0.72. Failure rate, by contrast, is essentially uncorrelated with all of them. That independence matters, because it says risk is not simply another face of low engagement, and it is part of why risky users are detected with an explicit rule rather than expected to fall out of a clustering of engagement variables.

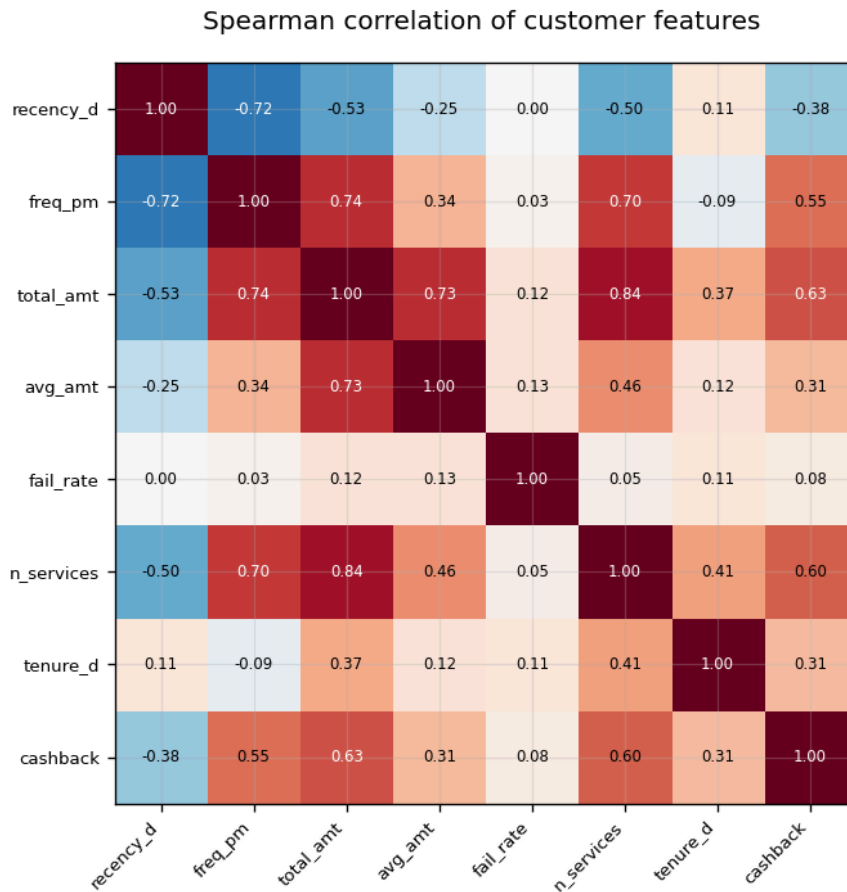


Figure 13. Rank correlations among customer features. Engagement variables are tightly linked; failure rate stands apart from all of them.

Principal component analysis compresses the same features into independent axes. The first component alone carries 55 percent of the variance and the first two carry 73 percent. The first component is an engagement-and-value axis, loading positively on frequency, spend and service diversity and negatively on recency, and customers flagged as churned separate cleanly along it even though churn was never an input to the analysis. Two practical readings follow: most of the structure in customer behaviour is low-dimensional, which supports the compact feature set used for segmentation in the next section, and the churn definition drawn from the recency valley in Figure 3 re-emerges here independently.

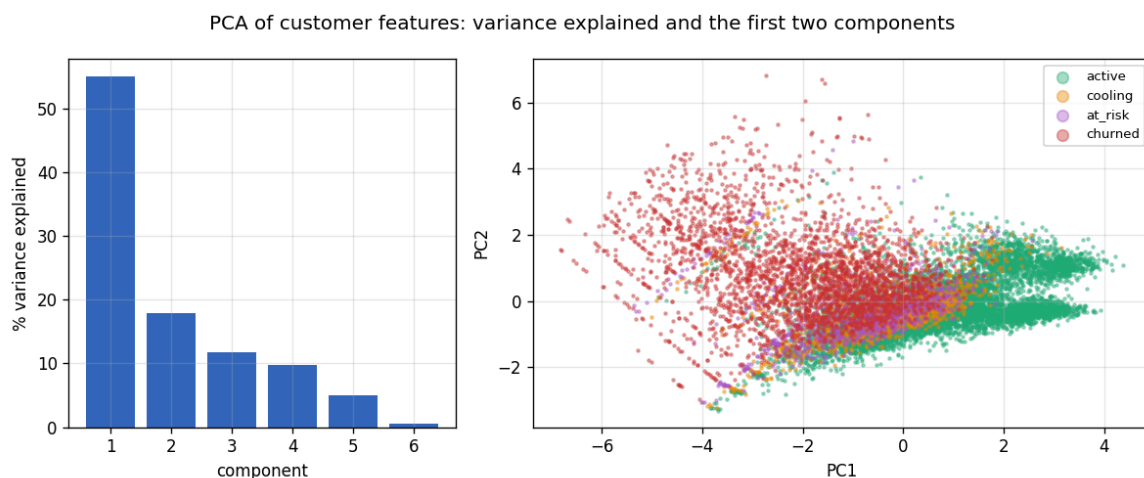


Figure 14. Variance explained by component, and the first two components with customers coloured by activity status. Churned customers separate along the first axis.

5. Customer segmentation

Segmentation was built on per-user features: recency, frequency and monetary value (the standard RFM triad), extended with failure rate, service diversity, cashback dependence and KYC status. For the clustering itself I used k-means, in its mini-batch form given one million rows. Density-based alternatives such as DBSCAN handle irregular cluster shapes well but behave poorly on heavy-tailed RFM data, where density varies enormously and much of the population ends up labelled as noise; hierarchical clustering needs quadratic memory and is simply infeasible at this scale. Skewed features were log-transformed first, because k-means measures Euclidean distance and would otherwise be dominated by the tails.

My first model did not survive its own validation, and I kept it in the notebook because the failure is instructive. It clustered on RFM plus failure rate plus a KYC indicator. The held-out labels revealed that the standardised binary KYC feature had captured an entire cluster by itself, and that the supposed risk cluster was really the population of users with one or two transactions, whose failure rates are extreme by arithmetic rather than behaviour (one failure in two attempts is fifty percent). The second model corrects both faults. It clusters on pure RFM, which is the structure k-means is actually good at finding, and detects risky users with an explicit rule that mirrors the brief's own definition: an elevated failure rate, shrunk by adding three phantom successes so that tiny histories cannot dominate, or failures combined with incomplete KYC. Statistical criteria preferred a coarser three-cluster solution (silhouette 0.47 against 0.40); I chose four clusters plus the rule overlay to match the personas the business needs, and state that trade-off rather than hiding it.

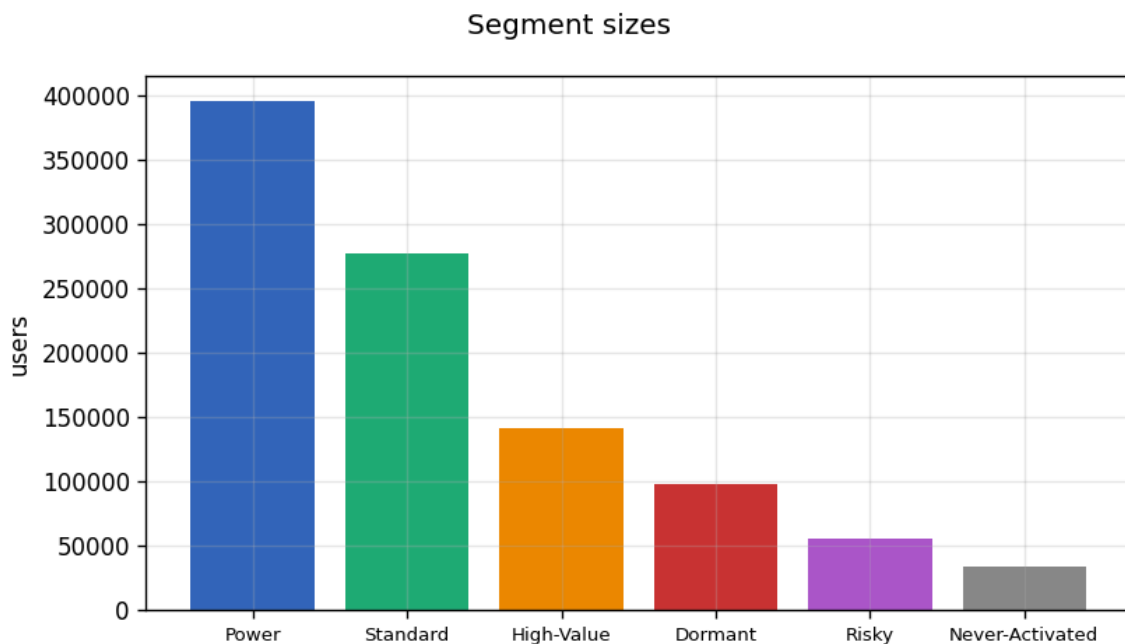


Figure 9. Users per segment, including signups that never made a transaction.

Segment	Users	Share	Recency (d)	Tx/month	Lifetime spend	Fail rate	KYC verified
High-Value	140,970	14.1%	2.7	10.4	Rs 315,049	4%	76%
Power	394,900	39.5%	22.1	1.9	Rs 50,878	5%	73%
Standard	277,322	27.7%	53.2	0.9	Rs 4,667	9%	78%
Dormant	97,967	9.8%	392.0	0.4	Rs 8,870	21%	74%
Risky	55,399	5.5%	148.3	1.1	Rs 45,851	48%	7%
Never-Activated	33,442	3.3%	-	0	0	-	-

Table 2. Segment profiles on the clean data. Each row is the behavioural identity card the targeting below is based on.

What defines each segment

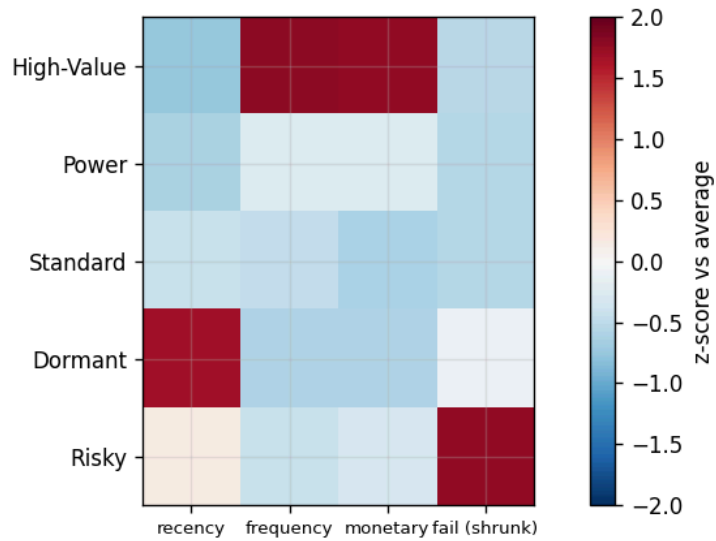


Figure 10. Segment centroids expressed as deviations from the average user. Each segment is defined by a distinct combination, not a single variable.

Segments in RFM space (15k sample)

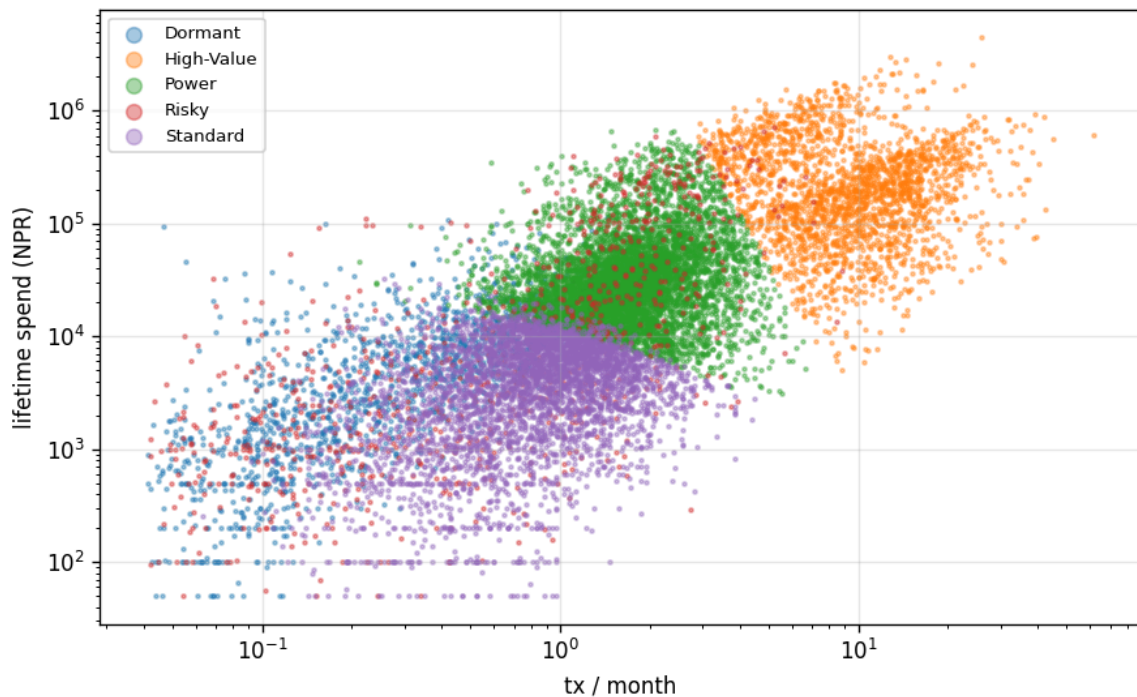


Figure 11. A 15,000-user sample in frequency and lifetime-spend space, log scales, coloured by segment.

The segments imply different actions. High-Value users are one in seven customers but carry the platform economically; a retention programme aimed at them (priority support, early access, service bundling) protects more value than an equivalent acquisition budget creates. Power users transact constantly at moderate value and are the natural base for referral and advocacy mechanics. Standard users' clearest upgrade path is a second product line, since service diversity separates them from the segments above. Dormant users are a re-engagement opportunity with a warning: their acquisition skews heavily toward cashback promotions, so any win-back offer should be tested against the possibility that they only ever came for the subsidy. Risky users, with a 48 percent failure rate and 7 percent verification, belong in a compliance review queue alongside the structuring accounts of Figure 8, and unverified-account limits are the immediate lever. Never-activated signups need an activation

nudge within days of registration, and Figure 4 says the one thing that nudge must guarantee is that the first attempted payment succeeds.

Because I generated the data, I also graded the segmentation against the held-out true labels rather than asking for trust. The recovered segments differ crisply in behaviour, which is what a business can act on. Where labels and clusters disagree, the disagreement traces to noise planted deliberately in the generator: a user labelled standard who behaves like a power user is, correctly, found among the engaged. Two limitations are worth naming. Frequency and spend overlap between the top two segments, so some High-Value and Power mass mixes; and the risk rule catches the failure-and-verification signature but not every irregular pattern. In production I would add sequence features such as burstiness and threshold proximity, and let the review queue's outcomes feed the rule.

6. Outlook

Monthly payment volume has grown from effectively zero at launch to Rs 6.1 billion in the last complete month, with total volume over the two years at Rs 64.7 billion and a 95.5 percent success rate. For planning purposes I extended the series three months forward using a plain linear trend fitted to the final year, and I label it a baseline rather than a forecast. With a series this short, dominated by user growth and carrying seasonality that is still emerging, an honest baseline communicates more than a sophisticated model would justify; a proper treatment with holdout backtesting is the natural next iteration and was consciously traded for depth in sections 4 and 5. The practical reading is straightforward: volume should be planned in the mid-single-digit billions per month and rising, cashback budgets scale with it, and the March incident already demonstrated what happens when capacity and reliability lag growth.

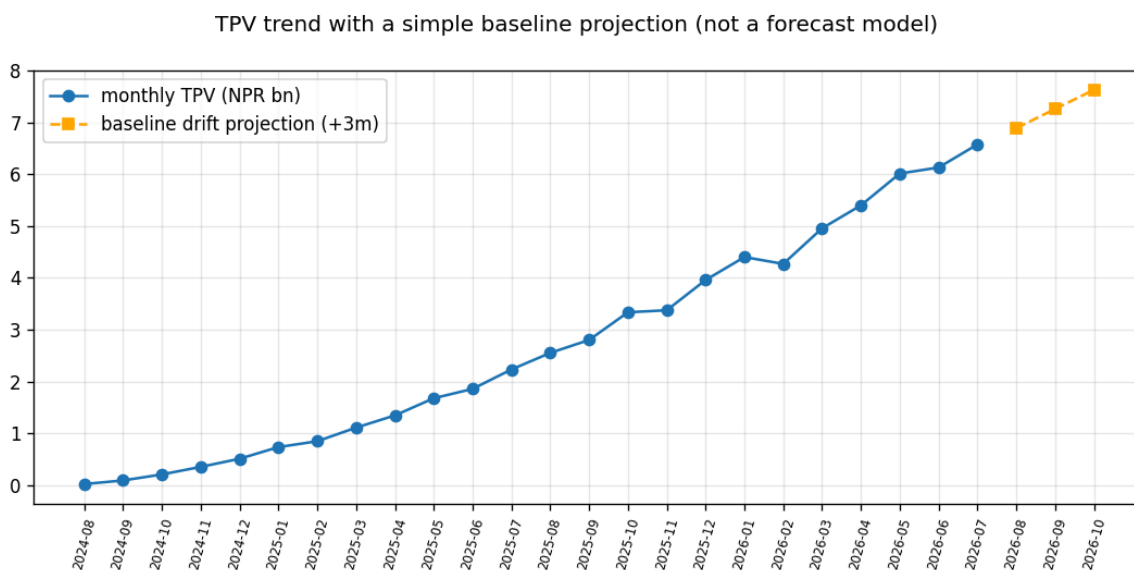


Figure 12. Monthly total payment volume with a three-month baseline projection, shown dashed. A baseline, deliberately not presented as a forecast model.

7. The dashboard

The interactive Power BI dashboard accompanies this report as a separate deliverable. It is built on aggregated extracts prepared in Python, so the workbook stays fast on ordinary hardware, and it is organised as four pages: an executive overview (volume, success rate, active users, churn, the trend of Figure 12), a growth and onboarding page (the funnel and the first-transaction retention gap), a risk and reliability page (Figures 4, 7 and 8 in interactive form, with the failure-reason mix), and a segment page (sizes, profiles, volume and churn by segment, verification status by segment). Slicers for month, province, referral source and segment run through

explicit dimension tables so a single control filters daily and monthly views coherently. One vocabulary choice is deliberate: the dashboard reports total payment volume, labelled TPV, and never calls it revenue. A wallet's revenue is a take-rate on that volume plus fees, neither of which this data models, and conflating the two would be the kind of silent overstatement the rest of this report tries to avoid.

8. Limitations

The dataset is synthetic. It is inspired by the product mix of a Nepali wallet but its numbers describe no real company, and the findings demonstrate method and judgement rather than facts about a market. Within that frame, the main analytical warnings have been stated where they arise: the first-failure effect on retention is causal here by construction and would need cohort evidence in production; the projection is a baseline, not a model; the segmentation blurs at the High-Value and Power boundary and its risk rule is a first pass; and payment volume is not revenue. I would rather present fewer claims with their weaknesses attached than more claims without them.

9. Reproducibility

Every number and figure in this report regenerates deterministically from fixed random seeds. The accompanying notebook builds the dataset, runs the audit, produces the features, charts and segmentation, and writes the extracts behind the dashboard; a reduced-scale setting reproduces every pattern in about two minutes for verification, and the full run completes in under twenty minutes on a modest laptop. The data files themselves are excluded from the submission by design, because the code is the dataset.